33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F16-SR-002
415V FLOTATION SWITCHROOM

F16-TR-005
TRANSFORMER ROOM

F16-TR-005
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

F16-SW-005
FLOTATION BLOWERS & COMPRESSORS 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
2.5VA
5P20
4000/ 5A
2.5VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
4000A
24VDC CONTROL POWER SUPPLY
NOTE 7
00F-08
125A
160A
SD
NOTE 6
415V 3NE~ 50Hz 4000A 65kA for 1sec
INC
CONTINUED ON SHT 2
2000-F16-SLD-EL-10212

01F-09	03F-09	4F-09	1R-09	3R-09	1F-01	3F-01	4F-01	1R-01	3R-01
800A	800A	800A	800A	xxxA	20A	20A	20A	20A	xxxA
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
100/1A	100/1A	100/1A	100/1A	xxVA	70A	70A	70A	70A	xxVA
EOL	EOL	EOL	EOL	EOL	EOL	EOL	EOL	EOL	EOL
RTD M 3~ 315kW	RTD M 3~ 315kW	RTD M 3~ 315kW	RTD M 3~ 315kW	RTD M 3~ 315kW	M 3~ 0.55kW	M 3~ 0.55kW	M 3~ 0.55kW	M 3~ 0.55kW	M 3~ 0.55kW
F16-BW-401 FLOTATION AIR BLOWER 1 2000-F16-SCM-EL-10255	F16-BW-402 FLOTATION AIR BLOWER 2 2000-F16-SCM-EL-10256	F16-BW-403 FLOTATION AIR BLOWER 3 2000-F16-SCM-EL-10257	F16-BW-404 FLOTATION AIR BLOWER 4 2000-F16-SCM-EL-10258	F16-BW-405 FLOTATION AIR BLOWER 5 2000-F21-SCM-EL-xxxxx (FUTURE)	F16-FN-487 FLOTATION AIR BLOWER 1 MOTOR COOLING FAN 2000-F16-SCM-EL-10247	F16-FN-488 FLOTATION AIR BLOWER 2 MOTOR COOLING FAN 2000-F16-SCM-EL-10248	F16-FN-489 FLOTATION AIR BLOWER 3 MOTOR COOLING FAN 2000-F16-SCM-EL-10249	F16-FN-490 FLOTATION AIR BLOWER 4 MOTOR COOLING FAN 2000-F16-SCM-EL-10250	F16-FN-505 FLOTATION AIR BLOWER 5 MOTOR COOLING FAN 2000-F21-SCM-EL-xxxxx (FUTURE)

E-SHUT
F16SR002-LCS
NOTE 12

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-005 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F16-SCM-EL-10254	F16-SW-005 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F16-SCM-EL-10253	F16-SW-005 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F16-SCM-EL-10251	F16-SW-005 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F16-DRG-EL-10101	F16-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F16-BLK-IC-10001	F16-SR-002 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F16-SLD-EL-10212	F16-SW-005 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	07SEP26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
1	26AUG25	FAT REDLINES INCORPORATED					
0	11JUN24	APPROVED FOR CONSTRUCTION					

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER AND SCAVENGER FLOTATION
F16-SW-005 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SLD-EL-10211	2

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